Integrating Demographic, Serological, Clinical, Environmental, and Spatiotemporal Determinants of Dengue Infection:

Towards a Predictive Framework for Early Risk Stratification and Hotspot Identification

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Presentation Outline

1. Background
2. Research Gap & Hypothesis
3. Research Objectives
4. Conceptual Framework
5. Statistical Methods for Predictive framework
6. Proposed Timeline
7. Conclusion
8. References

Background

- Dengue is a major mosquito-borne viral disease with substantial public health consequences, particularly in tropical and subtropical regions
- Approximately 390 million dengue virus infections occur annually, with around 5.6 billion people currently at risk of dengue and arboviral transmission
- Bangladesh has experienced substantial dengue transmission and recurrent outbreaks in recent years (World Health Organization, 2025a,b).
- Dengue infection is influenced by multiple interacting factors rather than a single determinant (Struchiner et al., 2015; Bhatt et al., 2013).



Figure 1: Aedes aegypti, dengue vector (**Source:** CDC image)

Research Gap & Hypothesis

Existing Fragmented Approaches

- Single-Domain Focus (Struchiner et al., 2015; Bhatt et al., 2013)
- Isolated Spatial/Temporal Analysis (Sanna et al., 2018; Morrison et al., 1998)
- Surveillance Delays (Bhatt et al., 2013; Struchiner et al., 2015)

Research Hypothesis

- H_1 **&** H_2**:** Multidomain data improve prediction of severe dengue compared with single-domain markers.
- H_3**:** Machine learning classifiers outperform traditional logistic regression in predictive performance.
- H_4**,** H_5 **&** H_6**:** Housing type influences dengue severity, with distinct seasonal patterns and significant spatiotemporal clustering.

Research Objectives

Primary Objective

The primary objective is to develop and evaluate an integrated predictive framework for dengue outcomes using demographic, serological, clinical, haematological, environmental, housing and geographic factors.

Specific Objectives

To fulfill primary objective this study consider some specific objectives. These are

- to identify individual-level demographic and socioeconomic determinants associated with dengue infection
- to investigate the association of serological and clinical characteristics with dengue infection and disease outcomes.
- to assess the influence of environmental and ecological conditions on dengue transmission.
- to investigate spatial, temporal and spatiotemporal patterns of dengue transmission.

Conceptual Framework

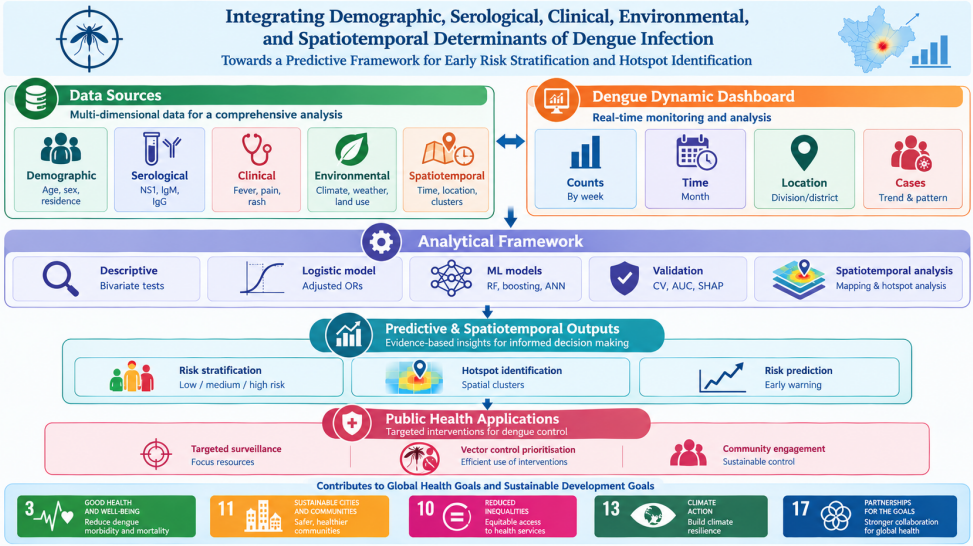


Figure 2: Integrated Framework for Dengue Risk Assessment and Surveillance

Statistical Methods for Predictive framework

Exploratory & Association

- Exploratory data analysis
- Descriptive & bivariate analysis
- Multivariable regression

Multilevel & Spatiotemporal

- Individual- and area-level effects
- Contextual & cross-level effects
- Spatial clustering
- Temporal & space-time patterns

Statistical & Machine Learning

- Logistic, GAM & Bayesian models
- Spatiotemporal models
- Random Forest, Boosting & ANN
- Ensemble learning & SHAP

Model Evaluation

Discrimination • Calibration • Cross-validation • Predictive accuracy

Proposed Research Timeline

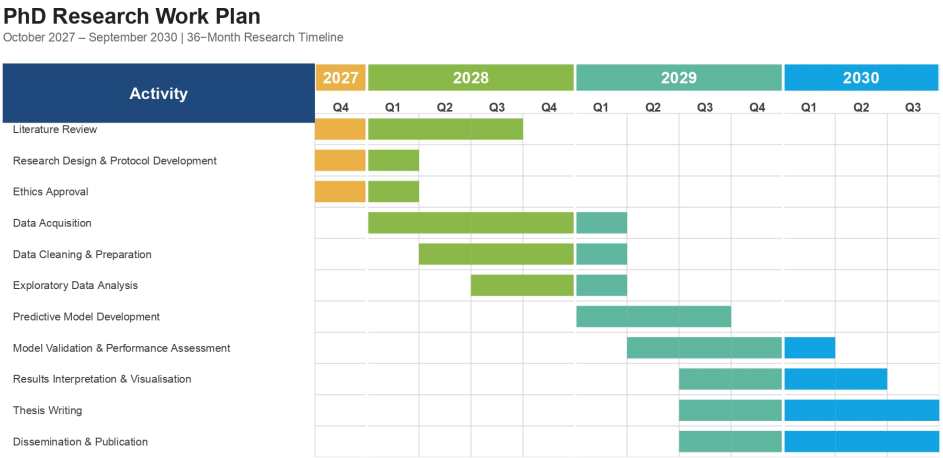


Figure 3: Proposed PhD research timeline and activity schedule (October 2027 – September 2030).

Conclusion

- Dengue transmission is a complex phenomenon involving interacting individual, environmental, spatial and temporal determinants.
- The proposed research will integrate these determinants within a unified statistical and epidemiological framework.
- Multilevel, spatial, spatiotemporal and machine-learning approaches will be combined to improve understanding and prediction of dengue risk.
- The findings are expected to contribute to stronger evidence-based dengue surveillance and prevention in Bangladesh.

References

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